Elementary Analysis

Chapter 1: Sections 1–3

All 28 exercises, with every subpart
Detailed worked solutions

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 1. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, letters in Section 3 denote real numbers. The empty sum is 0, and $0! = 1$.

Coverage. Section 1: Exercises 1.1–1.12. Section 2: Exercises 2.1–2.8. Section 3: Exercises 3.1–3.8.

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1 The set of natural numbers

Throughout this section, $\mathbb{N} = \{1, 2, 3, \ldots\}$. An induction proof has two separate obligations: establish an initial statement, and establish the implication from an arbitrary statement to the next one. In the second obligation, the induction hypothesis is an assumption used inside that implication; it is not a substitute for the initial case. When induction begins at an integer m instead of at 1, its conclusion concerns precisely the integers $n \geq m$.

We write $\sum_{k=r}^{s} u_k$ for the sum of the terms indexed by $r, r+1, \ldots, s$; if $r > s$, the sum is empty and equals 0. For integers $d \neq 0$ and M, saying that d divides M means that $M = dq$ for some integer q. The proofs of divisibility below identify this integer explicitly.

1.1 Exercise 1.1: Sum of the first squares

What we must show. For every $n \in \mathbb{N}$, establish

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

Plan. Increasing the upper limit from n to $n+1$ adds exactly the new term $(n+1)^2$. We substitute the proposed formula for the old sum, then factor the resulting polynomial into the expression required at $n+1$.

1. Specify the proposition and check the initial case.

Let P_n be the displayed identity. At $n = 1$, its left side is $1^2 = 1$, and its right side is

$$\frac{1(1+1)(2 \cdot 1 + 1)}{6} = \frac{1 \cdot 2 \cdot 3}{6} = 1.$$

Thus P_1 is true.

2. State the hypothesis and the next target.

Fix an arbitrary $n \in \mathbb{N}$ and assume P_n, namely

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

We must prove

$$\sum_{k=1}^{n+1} k^2 = \frac{(n+1)((n+1)+1)(2(n+1)+1)}{6} = \frac{(n+1)(n+2)(2n+3)}{6}.$$

3. Add the new square and carry out the factorization.

The induction hypothesis gives

$$\sum_{k=1}^{n+1} k^2 = \left(\sum_{k=1}^{n} k^2\right) + (n+1)^2$$

$$= \frac{n(n+1)(2n+1)}{6} + (n+1)^2$$
$$= \frac{n(n+1)(2n+1) + 6(n+1)^2}{6}$$
$$= \frac{(n+1)\big[n(2n+1) + 6(n+1)\big]}{6}.$$

To identify the bracket, expand its two products:

$$n(2n+1) + 6(n+1) = 2n^2 + n + 6n + 6 = 2n^2 + 7n + 6.$$

The needed factorization is verified by multiplying it back out:

$$(n+2)(2n+3) = 2n^2 + 3n + 4n + 6 = 2n^2 + 7n + 6.$$

Consequently,

$$\sum_{k=1}^{n+1} k^2 = \frac{(n+1)(n+2)(2n+3)}{6},$$

which is exactly P_{n+1}.

Conclusion. The initial case and the implication $P_n \Rightarrow P_{n+1}$ have both been proved. Mathematical induction therefore establishes the formula for every positive integer n.

1.2 Exercise 1.2: An arithmetic progression

What we must show. For every $n \in \mathbb{N}$, prove

$$\sum_{k=1}^{n} (8k - 5) = 4n^2 - n.$$

The summands at $k = 1$ and $k = 2$ are $8 - 5 = 3$ and $16 - 5 = 11$, so this notation describes the progression in the exercise.

Plan. Append its next term, $8(n+1) - 5$, and compare the result with the proposed expression evaluated at $n + 1$.

1. Initial case.

For $n = 1$, the left side is 3, and the right side is $4 \cdot 1^2 - 1 = 4 - 1 = 3$. Thus the statement holds at 1.

2. Induction hypothesis and target.

Let $n \geq 1$ and assume

$$\sum_{k=1}^{n} (8k - 5) = 4n^2 - n.$$

The required statement at the next index is

$$\sum_{k=1}^{n+1} (8k - 5) = 4(n+1)^2 - (n+1).$$

Before manipulating the sum, expand this target:

$$4(n+1)^2 - (n+1) = 4(n^2 + 2n + 1) - n - 1$$

$$= 4n^2 + 8n + 4 - n - 1$$
$$= 4n^2 + 7n + 3.$$

3. Compute the new sum.

The appended term is

$$8(n+1) - 5 = 8n + 8 - 5 = 8n + 3.$$

Using the induction hypothesis,

$$\begin{aligned}\sum_{k=1}^{n+1}(8k-5) &= \sum_{k=1}^{n}(8k-5) + \big(8(n+1)-5\big)\\ &= (4n^2 - n) + (8n + 3)\\ &= 4n^2 + (-n + 8n) + 3\\ &= 4n^2 + 7n + 3\\ &= 4(n+1)^2 - (n+1).\end{aligned}$$

This proves the next statement from the assumed one.

Conclusion. By induction, $3 + 11 + \cdots + (8n-5) = 4n^2 - n$ for all $n \in \mathbb{N}$.

1.3 Exercise 1.3: Sum of cubes and the square of a sum

What we must show. Prove, for every $n \in \mathbb{N}$, that

$$\sum_{k=1}^{n} k^3 = \left(\sum_{k=1}^{n} k\right)^2.$$

Plan. First record the formula for the sum inside the square. Then the asserted identity becomes a formula whose right side can be factored after the next cube is added. All square expansions used here follow from ordinary multiplication; no binomial theorem is being assumed.

1. Establish the needed formula for the first-power sum.

Write $T_n = 1 + 2 + \cdots + n$. We claim

$$T_n = \frac{n(n+1)}{2} \qquad (n \in \mathbb{N}).$$

For $n = 1$, both sides are 1. If $T_n = n(n+1)/2$ for some $n \geq 1$, then

$$\begin{aligned}T_{n+1} &= T_n + (n+1)\\ &= \frac{n(n+1)}{2} + \frac{2(n+1)}{2}\\ &= \frac{(n+1)(n+2)}{2}.\end{aligned}$$

This is the same formula with n replaced by $n+1$. Induction therefore proves the claim for every positive integer n.

It follows that the target identity can be written equivalently as

$$\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2 = \frac{n^2(n+1)^2}{4}.$$

2. Check the initial case for the cube identity.

For $n = 1$, the sum of cubes is $1^3 = 1$, while

$$\frac{1^2(1+1)^2}{4} = \frac{1 \cdot 4}{4} = 1.$$

Thus the proposed formula holds at 1.

3. State the induction hypothesis and next target.

Fix $n \geq 1$ and suppose

$$\sum_{k=1}^{n} k^3 = \frac{n^2(n+1)^2}{4}.$$

We must show

$$\sum_{k=1}^{n+1} k^3 = \frac{(n+1)^2(n+2)^2}{4}.$$

4. Append the next cube and factor.

Using the hypothesis and a common denominator,

$$\begin{aligned}
\sum_{k=1}^{n+1} k^3 &= \sum_{k=1}^{n} k^3 + (n+1)^3 \\
&= \frac{n^2(n+1)^2}{4} + (n+1)^3 \\
&= \frac{n^2(n+1)^2 + 4(n+1)^3}{4} \\
&= \frac{(n+1)^2\left[n^2 + 4(n+1)\right]}{4} \\
&= \frac{(n+1)^2(n^2+4n+4)}{4}.
\end{aligned}$$

Multiplication gives

$$(n+2)^2 = (n+2)(n+2) = n^2 + 2n + 2n + 4 = n^2 + 4n + 4.$$

Substituting this identity yields

$$\sum_{k=1}^{n+1} k^3 = \frac{(n+1)^2(n+2)^2}{4} = \left(\frac{(n+1)(n+2)}{2}\right)^2 = T_{n+1}^2.$$

This is precisely the assertion required at $n + 1$.

Conclusion. Mathematical induction proves $1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2$ for every positive integer n.

1.4 Exercise 1.4: The sum of consecutive odd numbers

What we must show. In part (a), use the first four sums to propose a formula. In part (b), prove the proposed formula for every positive integer n.

(a) Compute the requested examples and formulate the claim.

Let

$$O_n = 1 + 3 + \cdots + (2n-1) = \sum_{k=1}^{n} (2k-1).$$

The index k runs from 1 through n, so there are n summands. The requested calculations are

n	sum	O_n	n^2
1	1	1	1
2	$1+3$	4	4
3	$1+3+5$	9	9
4	$1+3+5+7$	16	16

These examples suggest $O_n = n^2$. Checking four values gives a candidate formula, but it does not establish what happens at every larger integer.

(b) Check the initial case.

At $n = 1$, $O_1 = 1 = 1^2$, so the formula starts correctly.

(b) State the induction hypothesis and next target.

Fix $n \geq 1$ and assume $O_n = n^2$. The target is

$$O_{n+1} = (n+1)^2.$$

The new last summand is the $(n+1)$st odd number:

$$2(n+1) - 1 = 2n + 2 - 1 = 2n + 1.$$

(b) Add the new odd number.

The hypothesis gives

$$\begin{aligned} O_{n+1} &= O_n + \big(2(n+1) - 1\big) \\ &= n^2 + (2n+1) \\ &= n^2 + 2n + 1 \\ &= (n+1)^2. \end{aligned}$$

For the last equality, expanding $(n+1)(n+1)$ gives $n^2 + n + n + 1 = n^2 + 2n + 1$.

Conclusion. The formula guessed in (a) is $1 + 3 + \cdots + (2n-1) = n^2$. Part (b) proves it for all $n \in \mathbb{N}$ by induction.

1.5 Exercise 1.5: A finite geometric sum

What we must show. For every positive integer n, prove

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = 2 - \frac{1}{2^n}.$$

There are $n+1$ terms on the left, since the leading 1 is 2^0. Equivalently, the sum is $G_n = \sum_{k=0}^{n} 2^{-k}$.

Plan. The next sum adds $1/2^{n+1}$. Expressing the old final fraction with denominator 2^{n+1} leaves precisely the new final fraction.

1. Initial case.

For $n = 1$, the left side is $1 + 1/2 = 3/2$, and the right side is $2 - 1/2 = 4/2 - 1/2 = 3/2$. Thus the required statement holds at 1.

2. Hypothesis and target.

Let $n \geq 1$ and assume $G_n = 2 - 1/2^n$. We must prove

$$G_{n+1} = 2 - \frac{1}{2^{n+1}}.$$

3. Add the next fraction with a common denominator.

Because $2^{n+1} = 2 \cdot 2^n$, we have $1/2^n = 2/2^{n+1}$. Hence

$$\begin{aligned} G_{n+1} &= G_n + \frac{1}{2^{n+1}} \\ &= 2 - \frac{1}{2^n} + \frac{1}{2^{n+1}} \\ &= 2 - \frac{2}{2^{n+1}} + \frac{1}{2^{n+1}} \\ &= 2 + \frac{-2+1}{2^{n+1}} \\ &= 2 - \frac{1}{2^{n+1}}. \end{aligned}$$

All denominators are nonzero because powers of 2 are positive.

Conclusion. The identity holds for every $n \in \mathbb{N}$ by mathematical induction.

1.6 Exercise 1.6: A difference of powers divisible by seven

What we must show. Show that, for every $n \in \mathbb{N}$, there is an integer q such that

$$11^n - 4^n = 7q.$$

Here both 11 and 4 are raised to the power n.

Plan. Express the next difference of powers as 11 times the old difference plus a term with a visible factor of 7.

1. Initial case and its integer witness.

For $n = 1$,

$$11^1 - 4^1 = 11 - 4 = 7 = 7 \cdot 1.$$

The witness is the integer $q = 1$, so 7 divides the difference at 1.

2. Induction hypothesis and next target.

Fix an arbitrary $n \geq 1$ and assume 7 divides $11^n - 4^n$. By the definition of divisibility, there is some $q \in \mathbb{Z}$ satisfying

$$11^n - 4^n = 7q.$$

We must produce an integer q' such that $11^{n+1} - 4^{n+1} = 7q'$.

3. Add and subtract a useful intermediate term.

Exponent rules give $11^{n+1} = 11 \cdot 11^n$ and $4^{n+1} = 4 \cdot 4^n$. Therefore

$$\begin{aligned} 11^{n+1} - 4^{n+1} &= 11 \cdot 11^n - 4 \cdot 4^n \\ &= 11 \cdot 11^n - 11 \cdot 4^n + 11 \cdot 4^n - 4 \cdot 4^n \\ &= 11(11^n - 4^n) + (11 - 4)4^n \\ &= 11(7q) + 7 \cdot 4^n \\ &= 7(11q + 4^n). \end{aligned}$$

The second line adds $-11 \cdot 4^n + 11 \cdot 4^n = 0$, so it leaves the quantity unchanged. Define $q' = 11q + 4^n$. Since q is an integer and 4^n is an integer, q' is an integer. The last displayed equality is the required divisibility witness at $n + 1$.

Conclusion. Induction proves that 7 divides $11^n - 4^n$ for every positive integer n.

1.7 Exercise 1.7: An exponential expression divisible by thirty-six

What we must show. Prove that 36 divides $7^n - 6n - 1$ for every $n \in \mathbb{N}$.

Plan. Multiplying the old expression by 7 nearly produces the next one. The remaining difference will be $36n$.

1. Initial case.

At $n = 1$,

$$7^1 - 6 \cdot 1 - 1 = 7 - 6 - 1 = 0 = 36 \cdot 0.$$

Thus the divisibility statement holds, with integer witness 0.

2. Hypothesis and next target.

Let $n \geq 1$ and assume the current statement. Then some $q \in \mathbb{Z}$ satisfies

$$7^n - 6n - 1 = 36q.$$

Our task is to express $7^{n+1} - 6(n + 1) - 1$ as 36 times an integer.

3. Relate the next expression to the old one.

First expand the expression at $n+1$:

$$7^{n+1} - 6(n+1) - 1 = 7 \cdot 7^n - 6n - 6 - 1 = 7 \cdot 7^n - 6n - 7.$$

On the other hand,

$$\begin{aligned} 7(7^n - 6n - 1) + 36n &= 7 \cdot 7^n - 42n - 7 + 36n \\ &= 7 \cdot 7^n + (-42 + 36)n - 7 \\ &= 7 \cdot 7^n - 6n - 7. \end{aligned}$$

These two computations establish the exact identity

$$7^{n+1} - 6(n+1) - 1 = 7(7^n - 6n - 1) + 36n.$$

Now insert the induction hypothesis:

$$\begin{aligned} 7^{n+1} - 6(n+1) - 1 &= 7(36q) + 36n \\ &= 36(7q + n). \end{aligned}$$

The number $7q + n$ is an integer because $q, n \in \mathbb{Z}$. It is therefore the needed witness for divisibility at $n+1$.

Conclusion. The base case and the induction step show that $36 \mid (7^n - 6n - 1)$ for every positive integer n.

1.8 Exercise 1.8: Induction with a later starting index

What we must show. Prove the inequality in (a) for all integers $n \geq 2$, and the inequality in (b) for all integers $n \geq 4$.

Plan. Use induction beginning at the first integer in each stated domain. In part (b), the inequality from (a) will supply the comparison needed after multiplying the induction hypothesis by $n+1$.

(a) Initial case for $n^2 > n + 1$.

At $n = 2$, the inequality is

$$2^2 = 4 > 3 = 2 + 1.$$

This gives the required base at the starting index 2.

(a) Induction hypothesis and next target.

Let n be an arbitrary integer with $n \geq 2$ and assume $n^2 > n + 1$. We must show

$$(n+1)^2 > (n+1) + 1 = n + 2.$$

Expand and then use the assumed strict inequality:

$$\begin{aligned} (n+1)^2 &= n^2 + 2n + 1 \\ &> (n+1) + 2n + 1 \\ &= 3n + 2. \end{aligned}$$

The inequality in the second line comes from adding the same number $2n + 1$ to both sides of $n^2 > n + 1$. Also,

$$(3n + 2) - (n + 2) = 2n \geq 4 > 0,$$

so $3n + 2 > n + 2$. Combining the two strict inequalities yields $(n + 1)^2 > n + 2$, as required.

Conclusion. Induction starting at 2 proves $n^2 > n + 1$ for every integer $n \geq 2$.

(b) Initial case for $n! > n^2$.

Recall that $n! = 1 \cdot 2 \cdots n$. At $n = 4$,

$$4! = 1 \cdot 2 \cdot 3 \cdot 4 = 24 > 16 = 4^2.$$

Thus the required base is true at 4.

(b) Induction hypothesis and next target.

Let $n \geq 4$ be an integer and assume $n! > n^2$. The next target is

$$(n + 1)! > (n + 1)^2.$$

The factorial definition gives $(n + 1)! = (n + 1)n!$. Since $n + 1 > 0$, multiplying the induction hypothesis by $n + 1$ preserves its direction:

$$(n + 1)! = (n + 1)n! > (n + 1)n^2.$$

Because $n \geq 4$ also implies $n \geq 2$, part (a) applies and gives $n^2 > n + 1$. Multiplying this inequality by the positive number $n + 1$ gives the second comparison

$$(n + 1)n^2 > (n + 1)(n + 1) = (n + 1)^2.$$

Putting the comparisons together,

$$(n + 1)! > (n + 1)n^2 > (n + 1)^2.$$

This is the needed induction step.

Conclusion. Induction beginning at 4 proves $n! > n^2$ for every integer $n \geq 4$. The different starting indices are essential: the conclusions of the two induction arguments are restricted to their stated domains.

1.9 Exercise 1.9: Exactly when an exponential exceeds a square

What we must show. Determine the integers n for which $2^n > n^2$, then justify the infinite portion of the answer by induction.

Plan. The strict inequality fails at some small positive integers, so an induction argument cannot begin at 1 and continue through every subsequent index. We check the exceptional small cases separately, begin the induction at 5, and also address zero and negative integers because the question says "integers."

(a) Exclude every negative integer.

Suppose $n \leq -1$ is an integer, and put $m = -n$. Then m is a positive integer and $n^2 = m^2 \geq 1$. By the definition of negative powers,

$$2^n = 2^{-m} = \frac{1}{2^m}.$$

The denominator 2^m is a product of $m \geq 1$ factors equal to 2, so $2^m \geq 2$. Since the denominator is positive, taking reciprocals gives $0 < 1/2^m \leq 1/2$. Hence

$$0 < 2^n \leq \frac{1}{2} < 1 \leq n^2.$$

In particular, $2^n > n^2$ is false for every negative integer n.

(a) Compute zero and the first five positive cases.

n	2^n	n^2	$2^n > n^2$?
0	1	0	yes
1	2	1	yes
2	4	4	no: equality
3	8	9	no: $8 < 9$
4	16	16	no: equality
5	32	25	yes

These calculations suggest that all integers at least 5 work. They also show that 0 and 1 are additional solutions, while $2, 3, 4$ are not.

(b) Set up the induction beginning at 5.

Let P_n denote $2^n > n^2$, now with the domain $n \geq 5$. The initial case is the checked inequality

$$2^5 = 32 > 25 = 5^2.$$

Fix an arbitrary integer $n \geq 5$ and assume $2^n > n^2$. We must prove

$$2^{n+1} > (n+1)^2.$$

Multiplication of the hypothesis by the positive number 2 gives

$$2^{n+1} = 2 \cdot 2^n > 2n^2.$$

To complete the step, we compare $2n^2$ with the next square.

(b) Verify the remaining polynomial inequality.

Expand the difference:

$$\begin{aligned} 2n^2 - (n+1)^2 &= 2n^2 - (n^2 + 2n + 1) \\ &= n^2 - 2n - 1 \\ &= n(n-2) - 1. \end{aligned}$$

Since $n \geq 5$, we have $n - 2 \geq 3 > 0$. Multiplication by $n - 2$ therefore preserves the inequality $n \geq 5$, and multiplication by 5 preserves $n - 2 \geq 3$. Thus

$$n(n-2) \geq 5(n-2) \geq 5 \cdot 3 = 15.$$

It follows that

$$2n^2 - (n+1)^2 = n(n-2) - 1 \geq 15 - 1 = 14 > 0.$$

Consequently $2n^2 > (n+1)^2$, and hence

$$2^{n+1} > 2n^2 > (n+1)^2.$$

This proves the implication $P_n \Rightarrow P_{n+1}$ for every integer $n \geq 5$. Induction now proves P_n for every such integer.

Conclusion. Taken literally over $\mathbb{Z}$, the complete solution set is

$$\{0, 1\} \ \cup \ \{n \in \mathbb{Z} : n \geq 5\}.$$

If "integers" is restricted to the positive integers in the surrounding discussion, the answer is $n = 1$ or $n \geq 5$. Zero is a solution whenever zero is included in the domain. The negative integers and $2, 3, 4$ have all been excluded, so no cases remain undecided.

1.10 Exercise 1.10: A sum whose lower and upper limits both change

What we must show. Show that for every positive integer n,

$$(2n+1) + (2n+3) + \cdots + (4n-1) = 3n^2.$$

Plan. Write the sum using an index before attempting induction. When n increases, the first old term disappears and two new terms enter. Accounting for all three changes is the central step.

1. Identify the summation limits and check the initial case.

Define

$$S_n = \sum_{k=n}^{2n-1} (2k+1).$$

At the lower limit $k = n$, the term is $2n+1$; at the upper limit $k = 2n-1$, it is $2(2n-1)+1 = 4n-1$. Consecutive values of k increase the term by 2. There are $(2n-1) - n + 1 = n$ terms, so this is exactly the sum in the question.

For $n = 1$, both limits equal 1. Thus

$$S_1 = 2 \cdot 1 + 1 = 3 = 3 \cdot 1^2.$$

The formula is true at its first index.

2. State the hypothesis and identify the next sum.

Fix $n \geq 1$ and assume $S_n = 3n^2$. We must prove $S_{n+1} = 3(n+1)^2$. Replacing n by $n+1$ in *both* limits gives

$$S_{n+1} = \sum_{k=n+1}^{2(n+1)-1} (2k+1) = \sum_{k=n+1}^{2n+1} (2k+1).$$

The old sum used the indices $n, n+1, \ldots, 2n-1$. The new sum uses $n+1, \ldots, 2n-1, 2n, 2n+1$. Therefore it removes the old index n and adds the indices $2n$ and $2n+1$.

3. Record the common portion, including the empty case.

Let

$$R_n = \sum_{k=n+1}^{2n-1} (2k+1).$$

For $n = 1$, this is the empty sum $\sum_{k=2}^{1}(2k+1) = 0$; the following decomposition remains valid in that case. For every $n \geq 1$,

$$\begin{aligned} S_n &= (2n+1) + R_n, \\ S_{n+1} &= R_n + \big(2(2n)+1\big) + \big(2(2n+1)+1\big) \\ &= R_n + (4n+1) + (4n+3). \end{aligned}$$

Solving the first equation for R_n and substituting it into the second establishes

$$S_{n+1} = S_n - (2n+1) + (4n+1) + (4n+3).$$

In particular, the change cannot be described merely by appending one term to S_n, since its lower limit also changes.

4. Substitute the hypothesis and simplify the new value.

$$\begin{aligned} S_{n+1} &= 3n^2 - (2n+1) + (4n+1) + (4n+3) \\ &= 3n^2 - 2n - 1 + 4n + 1 + 4n + 3 \\ &= 3n^2 + (-2+4+4)n + (-1+1+3) \\ &= 3n^2 + 6n + 3 \\ &= 3(n^2 + 2n + 1) \\ &= 3(n+1)^2. \end{aligned}$$

This is exactly the next assertion.

Conclusion. Induction proves the stated sum equals $3n^2$ for all $n \in \mathbb{N}$.

1.11 Exercise 1.11: A valid induction implication without a true base

What we must show. For P_n defined by "$n^2 + 5n + 1$ is an even integer," prove the implication requested in (a), determine its actual truth values in (b), and explain what this teaches about induction.

Plan. The difference between two consecutive expressions is even. That fact proves the requested implication but does not supply a single true initial statement. A separate parity calculation determines whether any initial statement could exist.

(a) Assume P_n and state what must follow.

Fix any $n \in \mathbb{N}$ and suppose P_n is true. An integer is even when it has the form $2q$ with $q \in \mathbb{Z}$, so the assumption says that

$$n^2 + 5n + 1 = 2q$$

for some integer q. We must prove $(n+1)^2 + 5(n+1) + 1$ is also twice an integer.

(a) Expand the next expression and give its witness.

Distributing the products gives

$$\begin{aligned}(n+1)^2 + 5(n+1) + 1 &= (n^2 + 2n + 1) + (5n + 5) + 1 \\ &= n^2 + 7n + 7 \\ &= (n^2 + 5n + 1) + (2n + 6) \\ &= (n^2 + 5n + 1) + 2(n + 3).\end{aligned}$$

Substitute the assumed representation:

$$(n+1)^2 + 5(n+1) + 1 = 2q + 2(n+3) = 2(q + n + 3).$$

Because $q + n + 3 \in \mathbb{Z}$, this expression is even. We have proved $P_n \Rightarrow P_{n+1}$ for every $n \in \mathbb{N}$.

(b) Determine the parity for an arbitrary n.

Rewrite the expression as

$$n^2 + 5n + 1 = n(n+5) + 1.$$

Every integer is either even or odd. Consider each case explicitly.

If n is even, write $n = 2r$ for some $r \in \mathbb{Z}$. Then

$$n(n+5) + 1 = 2r(n+5) + 1 = 2\big[r(n+5)\big] + 1.$$

The bracket is an integer, so this has the form $2\ell + 1$ and is odd.

If n is odd, write $n = 2r + 1$ for some $r \in \mathbb{Z}$. Now

$$n + 5 = 2r + 1 + 5 = 2r + 6 = 2(r + 3),$$

and hence

$$n(n+5) + 1 = n \cdot 2(r+3) + 1 = 2\big[n(r+3)\big] + 1.$$

Again the bracket is an integer, so the expression is odd.

Both cases show that $n^2 + 5n + 1$ is odd for every $n \in \mathbb{N}$. An odd integer cannot also be even: equality $2\ell + 1 = 2q$ would imply $1 = 2(q - \ell)$, whereas no integer multiplied by 2 equals 1. Thus P_n is false for every positive integer n.

(b) Explain the missing logical requirement.

The proposed initial statement P_1 is false because

$$1^2 + 5 \cdot 1 + 1 = 1 + 5 + 1 = 7,$$

and 7 is odd. Starting induction at a larger positive integer would not help, since every one of the proposed starting statements is false.

Part (a) is nevertheless a valid implication. It says that *if* one of these expressions were even, then the next would be even too. Its antecedent P_n is never true, so the implication has no true antecedent at which to start a chain of conclusions. In logical terms, the implication is *vacuously true*; it does not assert that either P_n or P_{n+1} is actually true.

Conclusion. There is no $n \in \mathbb{N}$ for which P_n is true, despite the validity of $P_n \Rightarrow P_{n+1}$ at every index. The moral is that induction requires both a true base case and the induction implication. Proving only the implication proves none of the assertions unconditionally.

1.12 Exercise 1.12: The binomial theorem from factorials and induction

What we must show. With

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \qquad 0 \le k \le n,$$

verify the first three cases, prove the identity in (b), and then prove

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k \qquad (n \in \mathbb{N},\ a, b \in \mathbb{R}).$$

Plan. Multiplication by $a+b$ produces two sums. After shifting the index in the sum multiplied by b, the terms with the same powers have adjacent coefficients. Part (b) combines those coefficients into the ones required for exponent $n+1$.

Conventions and endpoint coefficients.

For a positive integer r, $r! = 1 \cdot 2 \cdots r$, and $0! = 1$. All factorials used in denominators are therefore positive. In particular, for each positive integer n,

$$\binom{n}{0} = \frac{n!}{0!n!} = \frac{n!}{1 \cdot n!} = 1, \qquad \binom{n}{n} = \frac{n!}{n!0!} = \frac{n!}{n! \cdot 1} = 1.$$

The endpoint coefficients will be treated separately from the interior ones, so no factorial of a negative integer will be needed.

For the polynomial expressions here, a power with exponent 0 is an empty product and denotes the constant 1. Thus the expression x^0 in a monomial has value 1 even when the value substituted for x is 0. This convention identifies the endpoint monomials as a^n and b^n. No step below divides by a or by b, so zero values of either variable are permitted.

(a) Verify the theorem at $n = 1$.

Here $1! = 1$ and $0! = 1$, so

$$\binom{1}{0} = \frac{1!}{0!1!} = 1, \qquad \binom{1}{1} = \frac{1!}{1!0!} = 1.$$

The proposed right side is

$$\binom{1}{0} a^1 b^0 + \binom{1}{1} a^0 b^1 = 1 \cdot a \cdot 1 + 1 \cdot 1 \cdot b = a + b,$$

which equals $(a+b)^1$. The identity therefore holds for every pair of real numbers a, b at $n = 1$.

(a) Verify the theorem at $n = 2$ by multiplication.

Since $2! = 2$, the coefficients are

$$\binom{2}{0} = \frac{2}{1 \cdot 2} = 1, \qquad \binom{2}{1} = \frac{2}{1 \cdot 1} = 2, \qquad \binom{2}{2} = \frac{2}{2 \cdot 1} = 1.$$

The distributive law and commutativity of real multiplication give

$$(a+b)^2 = (a+b)(a+b)$$

$$
\begin{aligned}
&= a(a+b) + b(a+b) \\
&= a^2 + ab + ba + b^2 \\
&= a^2 + ab + ab + b^2 \\
&= a^2 + 2ab + b^2.
\end{aligned}
$$

Using the coefficients just computed, this is

$$\binom{2}{0}a^2b^0 + \binom{2}{1}a^1b^1 + \binom{2}{2}a^0b^2,$$

as required. This calculation does not appeal to a binomial expansion; it establishes this particular expansion directly.

(a) Verify the theorem at $n = 3$ by another multiplication.

The relevant factorials are $3! = 6$, $2! = 2$, $1! = 1$, and $0! = 1$. Consequently

$$\binom{3}{0} = \frac{6}{1 \cdot 6} = 1, \quad \binom{3}{1} = \frac{6}{1 \cdot 2} = 3, \quad \binom{3}{2} = \frac{6}{2 \cdot 1} = 3, \quad \binom{3}{3} = \frac{6}{6 \cdot 1} = 1.$$

Multiply the identity already established for the square by $a + b$:

$$
\begin{aligned}
(a+b)^3 &= (a+b)(a^2 + 2ab + b^2) \\
&= a(a^2 + 2ab + b^2) + b(a^2 + 2ab + b^2) \\
&= a^3 + 2a^2b + ab^2 + a^2b + 2ab^2 + b^3 \\
&= a^3 + (2+1)a^2b + (1+2)ab^2 + b^3 \\
&= a^3 + 3a^2b + 3ab^2 + b^3.
\end{aligned}
$$

The coefficients are exactly $\binom{3}{0}, \binom{3}{1}, \binom{3}{2}, \binom{3}{3}$, so this is the asserted formula at $n = 3$, for all real a, b.

(b) Set the range of the indices and obtain a common denominator.

Fix $n \in \mathbb{N}$ and an integer k with $1 \le k \le n$. We must prove

$$\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}.$$

The bounds on k give $k - 1 \ge 0$ and $n - k \ge 0$, so every factorial in the two coefficients is defined. From the factorial definition,

$$(n-k+1)! = (n-k+1)(n-k)!, \qquad k! = k(k-1)!.$$

The numbers k and $n-k+1$ are both positive. Therefore the following rewritings use nonzero factors and are valid:

$$
\begin{aligned}
\binom{n}{k} &= \frac{n!}{k!(n-k)!} = \frac{n!(n-k+1)}{k!(n-k+1)!}, \\
\binom{n}{k-1} &= \frac{n!}{(k-1)!(n-(k-1))!} = \frac{n!}{(k-1)!(n-k+1)!} = \frac{n!k}{k!(n-k+1)!}.
\end{aligned}
$$

(b) Add the numerators and identify the next coefficient.

Both fractions now have denominator $k!(n-k+1)!$, and hence

$$\begin{aligned}\binom{n}{k}+\binom{n}{k-1} &= \frac{n!(n-k+1)+n!k}{k!(n-k+1)!}\\ &= \frac{n!\big[(n-k+1)+k\big]}{k!(n-k+1)!}\\ &= \frac{n!(n+1)}{k!(n-k+1)!}\\ &= \frac{(n+1)!}{k!((n+1)-k)!}\\ &= \binom{n+1}{k}.\end{aligned}$$

The penultimate line uses $(n+1)! = (n+1)n!$ and $n-k+1 = (n+1)-k$. This proves the claimed identity for every integer k in the stipulated range $1,\dots,n$. It is often called Pascal's identity. This derivation uses only the definition of the coefficients, not the binomial theorem being proved.

(c) State the induction proposition and verify its base.

Let P_n be the statement that

$$(a+b)^n = \sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k \quad \text{for every } a,b\in\mathbb{R}.$$

Part (a) established P_1: the two terms in that sum are a and b. Thus this proposition has a valid starting case. The checks at 2 and 3 in (a) are consistent with the result but are not needed as additional bases for induction.

(c) State the induction hypothesis and the complete next target.

Let $n \geq 1$ be arbitrary and assume P_n. Fix arbitrary real numbers a and b. The hypothesis gives

$$(a+b)^n = \sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k$$

for these particular numbers. We must prove

$$(a+b)^{n+1} = \sum_{k=0}^{n+1}\binom{n+1}{k}a^{n+1-k}b^k.$$

Because a and b were chosen arbitrarily, obtaining this equality will prove the assertion for every pair of real numbers, as P_{n+1} requires.

(c) Multiply the assumed identity and distribute both sums.

Multiply the induction hypothesis by $a+b$. Distributivity of finite sums gives

$$\begin{aligned}(a+b)^{n+1} &= (a+b)(a+b)^n\\ &= (a+b)\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k\end{aligned}$$

$$= a\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k + b\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k$$
$$= \sum_{k=0}^{n}\binom{n}{k}a^{n+1-k}b^k + \sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^{k+1}.$$

In the first sum, multiplying a^{n-k} by a raises its exponent to $n-k+1 = n+1-k$. In the second sum, multiplying b^k by b raises its exponent to $k+1$. All of these are products of nonnegative integer powers, so these rules are valid at $a = 0$ and $b = 0$ as well.

(c) Reindex the second sum with its new bounds.

In that second sum, set $j = k+1$. Then $k = j-1$, and the bounds change as follows:

$$k = 0 \implies j = 1, \qquad k = n \implies j = n+1.$$

Also the power of a becomes

$$n - k = n - (j-1) = n + 1 - j.$$

Every old index corresponds to exactly one new index and conversely. Therefore

$$\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^{k+1} = \sum_{j=1}^{n+1}\binom{n}{j-1}a^{n+1-j}b^j.$$

Substituting the reindexed sum into the preceding expression gives

$$(a+b)^{n+1} = \sum_{k=0}^{n}\binom{n}{k}a^{n+1-k}b^k + \sum_{j=1}^{n+1}\binom{n}{j-1}a^{n+1-j}b^j.$$

(c) Separate the two endpoints before combining interior terms.

The first sum contains an endpoint at $k = 0$, equal to

$$\binom{n}{0}a^{n+1}b^0 = 1 \cdot a^{n+1} \cdot 1 = a^{n+1}.$$

The second sum contains an endpoint at $j = n+1$, equal to

$$\binom{n}{n}a^0b^{n+1} = 1 \cdot 1 \cdot b^{n+1} = b^{n+1}.$$

Removing these endpoints leaves the common index range $1, \ldots, n$. Rename the dummy index j in the second interior sum as k; this changes only the name of the summation variable. We obtain

$$(a+b)^{n+1} = a^{n+1} + \sum_{k=1}^{n}\binom{n}{k}a^{n+1-k}b^k + \sum_{k=1}^{n}\binom{n}{k-1}a^{n+1-k}b^k + b^{n+1}$$
$$= a^{n+1} + \sum_{k=1}^{n}\left(\binom{n}{k} + \binom{n}{k-1}\right)a^{n+1-k}b^k + b^{n+1}.$$

For each interior index k, both terms contain the identical monomial $a^{n+1-k}b^k$; the distributive law combines their coefficients.

(c) Apply part (b) and restore the full target sum.

Every index in the remaining sum satisfies $1 \le k \le n$, so part (b) applies to every coefficient. Thus

$$(a+b)^{n+1} = a^{n+1} + \sum_{k=1}^{n} \binom{n+1}{k} a^{n+1-k} b^k + b^{n+1}.$$

The endpoint computations at the beginning of the exercise, applied with $n+1$ in place of n, give

$$\binom{n+1}{0} = 1, \qquad \binom{n+1}{n+1} = 1.$$

Consequently the first and last terms can be written as

$$a^{n+1} = \binom{n+1}{0} a^{n+1} b^0, \qquad b^{n+1} = \binom{n+1}{n+1} a^0 b^{n+1}.$$

Including these two endpoint terms with the interior sum produces

$$(a+b)^{n+1} = \sum_{k=0}^{n+1} \binom{n+1}{k} a^{n+1-k} b^k,$$

which is exactly the required next statement for the arbitrary real numbers a, b. Therefore P_n implies P_{n+1}.

Conclusion. Part (a) verifies the first three cases, part (b) establishes Pascal's identity directly from factorials, and part (c) proves the binomial theorem for every positive integer n and every $a, b \in \mathbb{R}$ by mathematical induction. The proof includes $a = 0$, $b = 0$, and $a = b = 0$, since it uses no division by either variable and interprets zero exponents as constant monomials.

2 The set of rational numbers

The exercises in this section ask whether particular numbers belong to $\mathbb{Q}$. Recall that a number is rational when it can be written as m/n, where $m, n \in \mathbb{Z}$ and $n \neq 0$. A number is irrational when it is a real number that is not rational. The following elementary tools explain the steps used in the solutions.

Tool 1: Why rational arithmetic stays rational

Suppose $r = m/n$ and $s = p/q$, where $m, n, p, q \in \mathbb{Z}$ and $n, q \neq 0$. Putting fractions over a common denominator gives

$$r + s = \frac{mq + pn}{nq}, \qquad r - s = \frac{mq - pn}{nq}, \qquad rs = \frac{mp}{nq}.$$

Each numerator is an integer, and nq is a nonzero integer. Each displayed quantity is therefore rational by the definition of $\mathbb{Q}$. If $s \neq 0$, then $p \neq 0$, and

$$\frac{r}{s} = \frac{mq}{np}$$

also has an integer numerator and a nonzero integer denominator. Thus division by a nonzero rational number preserves rationality. Integers are rational because $j = j/1$. In particular, multiplication by an integer and subtraction of an integer preserve rationality.

For a positive integer k, repeated multiplication gives

$$r^k = \underbrace{r \cdot r \cdots r}_{k \text{ factors}} = \frac{m^k}{n^k} \in \mathbb{Q},$$

since $m^k, n^k \in \mathbb{Z}$ and $n^k \neq 0$. These formulas justify every use of "rational closure" below.

Tool 2: The integer cancellation fact behind the rational zeros theorem

Two integers are relatively prime when they have no common positive divisor other than 1. We use this elementary fact: if $u, v \in \mathbb{Z}$ are relatively prime, $v \neq 0$, and v divides uw, then v divides w. Here "v divides z" means that $z = vk$ for some $k \in \mathbb{Z}$.

For completeness, the cancellation fact follows from a short integer argument. The positive integers of the form $Au + Bv$, with $A, B \in \mathbb{Z}$, form a nonempty set: one of v and $-v$ is positive. Let d be the least such positive integer, and write $d = A_0u + B_0v$. The division algorithm gives $u = hd + t$, with $h \in \mathbb{Z}$ and $0 \leq t < d$. Then

$$t = u - hd = (1 - hA_0)u - hB_0v.$$

If $t > 0$, this would be a smaller positive integer of the same form, contradicting the choice of d. Hence $t = 0$, so d divides u. Applying the same argument to division of v by d shows that d divides v. Relative primality forces $d = 1$. Consequently there are integers A_0, B_0 with

$$A_0u + B_0v = 1.$$

Multiplying this identity by w gives

$$w = A_0(uw) + B_0vw.$$

If v divides uw, both terms on the right are integer multiples of v, so w is also an integer multiple of v. This proves the cancellation fact. The integer division algorithm and the existence of a least positive integer are the only integer facts used here.

One consequence needed below is worth spelling out. If u, v are relatively prime and v divides u^k, then cancellation in $u^k = u\,u^{k-1}$ shows that v divides u^{k-1}. Repeating this step eventually shows that v divides 1. When $u \neq 0$, the last power in this argument is $u^0 = 1$. When $u = 0$, relative primality already forces $|v| = 1$, so the same conclusion holds directly.

Tool 3: A monic polynomial can have only integer rational zeros

A polynomial is *monic* when the coefficient of its highest power is 1. Suppose

$$P(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0, \qquad a_0, \ldots, a_{d-1} \in \mathbb{Z},$$

where $d \geq 1$. Assume that $r \in \mathbb{Q}$ and $P(r) = 0$. Write $r = m/n$ in lowest terms, with $n > 0$. Thus $m, n \in \mathbb{Z}$ are relatively prime. If $m = 0$, then $r = 0$ is already an integer. For the denominator argument we may therefore assume $m \neq 0$. Substituting and multiplying by n^d gives

$$m^d + \sum_{j=0}^{d-1} a_j m^j n^{d-j} = 0.$$

Every term in the sum contains a factor n. Equivalently,

$$m^d = -n \left(\sum_{j=0}^{d-1} a_j m^j n^{d-j-1} \right).$$

The expression in parentheses is an integer, so n divides m^d. Tool 2, applied repeatedly to the relatively prime integers m, n, shows that n divides 1. As n is positive, $n = 1$. Therefore $r = m \in \mathbb{Z}$.

There is a further restriction if $a_0 \neq 0$. In that case $r \neq 0$, because $P(0) = a_0 \neq 0$. Substituting the integer m into $P(m) = 0$ and collecting all terms other than the constant gives

$$a_0 = -m \left(m^{d-1} + \sum_{j=1}^{d-1} a_j m^{j-1} \right).$$

For $d = 1$ the sum is empty and equals 0. In every case the expression in parentheses is an integer. Hence m divides a_0.

We have proved the monic case of Ross's Rational Zeros Theorem (Theorem 2.2 and Corollary 2.3):

> A rational zero of a monic polynomial with integer coefficients must be an integer. If the constant coefficient is nonzero, that integer must divide the constant coefficient.

In particular, if N is a positive integer and k is a positive integer, then a rational solution of $x^k = N$ must be an integer, because it is a zero of the monic integer polynomial $x^k - N$.

Tool 4: Bounding a positive root by consecutive integers

For $0 \leq u < v$ and a positive integer k,

$$v^k - u^k = (v - u)(v^{k-1} + v^{k-2}u + \cdots + u^{k-1}) > 0.$$

For $k = 1$ the factor in parentheses is 1. For $k \geq 2$, all its terms are nonnegative and its first term v^{k-1} is positive. Thus raising nonnegative numbers to the kth power preserves their strict order. This is an algebraic fact; no limiting argument is needed.

Let x be the nonnegative kth root of a positive integer N. Suppose a nonnegative integer a satisfies

$$a^k < N < (a+1)^k.$$

If $x \leq a$, then $x^k \leq a^k < N$, contradicting $x^k = N$. If $x \geq a+1$, then $x^k \geq (a+1)^k > N$, giving the same contradiction. It follows that

$$a < x < a+1.$$

There is no integer strictly between the consecutive integers a and $a+1$. Tool 3 therefore implies that x is irrational. This gives the criterion used in the first two exercises:

> If a positive integer lies strictly between two consecutive nonnegative integer kth powers, its positive kth root is irrational.

2.1 Exercise 2.1: Five square roots of integers

What we must show. Prove that $\sqrt{3}$, $\sqrt{5}$, $\sqrt{7}$, $\sqrt{24}$, and $\sqrt{31}$ are irrational.

Plan. Each number is a root of a monic integer polynomial $x^2 - N$. If it were rational, Tool 3 would make it an integer. We locate each root strictly between consecutive integers to rule that out.

1. Record the exact integer-square comparisons

The needed computations are

$$1^2 = 1, \quad 2^2 = 4, \quad 3^2 = 9, \quad 4^2 = 16, \quad 5^2 = 25, \quad 6^2 = 36.$$

They give the following bounds by Tool 4:

N	Consecutive square bounds	Consequent root bounds
3	$1^2 = 1 < 3 < 4 = 2^2$	$1 < \sqrt{3} < 2$
5	$2^2 = 4 < 5 < 9 = 3^2$	$2 < \sqrt{5} < 3$
7	$2^2 = 4 < 7 < 9 = 3^2$	$2 < \sqrt{7} < 3$
24	$4^2 = 16 < 24 < 25 = 5^2$	$4 < \sqrt{24} < 5$
31	$5^2 = 25 < 31 < 36 = 6^2$	$5 < \sqrt{31} < 6$

2. Apply the rational-root argument to each number

(a) Suppose $\sqrt{3}$ were rational. Since $(\sqrt{3})^2 - 3 = 0$, Tool 3 applied to $x^2 - 3$ would imply $\sqrt{3} \in \mathbb{Z}$. But $1 < \sqrt{3} < 2$, and no integer satisfies those two inequalities. This contradiction proves $\sqrt{3} \notin \mathbb{Q}$.

(b) Suppose $\sqrt{5}$ were rational. The equation $(\sqrt{5})^2 - 5 = 0$ and Tool 3 imply $\sqrt{5} \in \mathbb{Z}$. The bounds $2 < \sqrt{5} < 3$ exclude every integer, so $\sqrt{5} \notin \mathbb{Q}$.

(c) Suppose $\sqrt{7}$ were rational. The monic integer equation $(\sqrt{7})^2 - 7 = 0$ would make $\sqrt{7}$ an integer. The strict bounds $2 < \sqrt{7} < 3$ make this impossible. Hence $\sqrt{7} \notin \mathbb{Q}$.

(d) Suppose $\sqrt{24}$ were rational. Because it satisfies $x^2 - 24 = 0$, it would be an integer by Tool 3. Its bounds $4 < \sqrt{24} < 5$ contradict that conclusion. Thus $\sqrt{24} \notin \mathbb{Q}$.

(e) Suppose $\sqrt{31}$ were rational. Tool 3 applied to x^2-31 would make it an integer. Since $5 < \sqrt{31} < 6$, there is no such integer. Therefore $\sqrt{31} \notin \mathbb{Q}$.

3. Establish one auxiliary case for later exercises

We will also need that $\sqrt{2}$ is irrational. If it were rational, the equation $(\sqrt{2})^2 - 2 = 0$ would make it an integer. But

$$1^2 = 1 < 2 < 4 = 2^2 \quad \Longrightarrow \quad 1 < \sqrt{2} < 2,$$

which is impossible for an integer. Thus $\sqrt{2} \notin \mathbb{Q}$ as well.

Conclusion. All five requested square roots are irrational. The same argument also proves the auxiliary fact $\sqrt{2} \notin \mathbb{Q}$.

2.2 Exercise 2.2: Three higher roots of integers

What we must show. Prove that $\sqrt[3]{2}$, $\sqrt[7]{5}$, and $\sqrt[4]{13}$ are irrational.

Plan. Tool 3 applies to $x^3 - 2$, $x^7 - 5$, and $x^4 - 13$. We compute suitable consecutive integer powers, then apply Tool 4.

1. The cube root of 2

Let $x = \sqrt[3]{2}$. By the definition of the positive cube root, $x > 0$ and $x^3 = 2$. Assume that $x \in \mathbb{Q}$. Since x is a zero of the monic polynomial $x^3 - 2$, whose coefficients are integers, Tool 3 gives $x \in \mathbb{Z}$.

Now $1^3 = 1$ and $2^3 = 2 \cdot 2 \cdot 2 = 8$. Consequently

$$1^3 = 1 < 2 < 8 = 2^3.$$

Tool 4 gives $1 < x < 2$. No integer lies in that interval, contradicting $x \in \mathbb{Z}$. Hence $\sqrt[3]{2}$ is irrational.

2. The seventh root of 5

Let $y = \sqrt[7]{5}$. Then $y > 0$ and $y^7 = 5$. If y were rational, Tool 3 applied to $x^7 - 5$ would imply $y \in \mathbb{Z}$. The exact powers are

$$1^7 = 1, \qquad 2^7 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 128.$$

Thus

$$1^7 = 1 < 5 < 128 = 2^7,$$

so $1 < y < 2$. This excludes all integers and contradicts the consequence of the rationality assumption. Therefore $\sqrt[7]{5}$ is irrational.

3. The fourth root of 13

Let $z = \sqrt[4]{13}$, where the fourth-root notation denotes the positive real root. Then $z > 0$ and $z^4 = 13$. If $z \in \mathbb{Q}$, the monic integer equation $z^4 - 13 = 0$ gives $z \in \mathbb{Z}$ by Tool 3. However,

$$1^4 = 1 < 13 < 16 = 2 \cdot 2 \cdot 2 \cdot 2 = 2^4.$$

It follows that $1 < z < 2$, contradicting integer membership. Hence $\sqrt[4]{13}$ is irrational.

Conclusion. The numbers $\sqrt[3]{2}$, $\sqrt[7]{5}$, and $\sqrt[4]{13}$ are all irrational.

2.3 Exercise 2.3: A nested square root

What we must show. Prove that $\sqrt{2+\sqrt{2}}$ is irrational.

Plan. Assume the outer square root is rational. Square it and isolate $\sqrt{2}$. Rational closure would then force $\sqrt{2}$ to be rational, contradicting the auxiliary case in Exercise 2.1.

1. Name the number and record its defining equation

Put

$$a = \sqrt{2+\sqrt{2}}.$$

The inner root $\sqrt{2}$ is positive, so $2+\sqrt{2} > 0$. The outer square root therefore denotes a positive real number. Its defining equation is

$$a^2 = 2+\sqrt{2}.$$

2. Assume rationality and rearrange

Suppose, for a contradiction, that $a \in \mathbb{Q}$. Then $a^2 \in \mathbb{Q}$ by rational closure under multiplication. Subtracting the rational integer 2 from both sides gives

$$\sqrt{2} = a^2 - 2 \in \mathbb{Q}.$$

In fraction form, if $a = m/n$ with $m, n \in \mathbb{Z}$ and $n \neq 0$, this is

$$a^2 - 2 = \frac{m^2}{n^2} - \frac{2n^2}{n^2} = \frac{m^2-2n^2}{n^2},$$

an integer divided by a nonzero integer. Thus the rationality conclusion is justified directly by the definition, as well as by Tool 1.

3. Identify the contradiction

Exercise 2.1 established $\sqrt{2} \notin \mathbb{Q}$. The rationality assumption for a has produced its negation, $\sqrt{2} \in \mathbb{Q}$. Therefore the assumption is false.

Conclusion. $\sqrt{2+\sqrt{2}}$ is irrational.

2.4 Exercise 2.4: A nested cube root

What we must show. Prove that $\sqrt[3]{5-\sqrt{3}}$ is irrational.

Plan. Cube the number to remove the outer radical, then isolate $\sqrt{3}$. If the original number were rational, this rearrangement would express $\sqrt{3}$ using rational operations only.

1. Check the sign and write the cube equation

Exercise 2.1 gives $1 < \sqrt{3} < 2$. In particular,

$$5 - \sqrt{3} > 5 - 2 = 3 > 0.$$

Thus

$$a = \sqrt[3]{5-\sqrt{3}}$$

is a positive real number, and the definition of cube root gives

$$a^3 = 5 - \sqrt{3}.$$

2. Derive a rational expression for $\sqrt{3}$ under the assumption

Assume $a \in \mathbb{Q}$. Then $a^3 \in \mathbb{Q}$ by Tool 1. Adding $\sqrt{3}$ to the cube equation and subtracting a^3 gives

$$a^3 + \sqrt{3} = 5, \qquad \text{hence} \qquad \sqrt{3} = 5 - a^3.$$

Both 5 and a^3 are rational, so their difference is rational. Explicitly, if $a = m/n$ with integer m, n and $n \neq 0$, then

$$5 - a^3 = \frac{5n^3 - m^3}{n^3} \in \mathbb{Q}.$$

3. Finish the contradiction

The preceding step gives $\sqrt{3} \in \mathbb{Q}$, contrary to Exercise 2.1. Therefore $a \notin \mathbb{Q}$.

Conclusion. $\sqrt[3]{5 - \sqrt{3}}$ is irrational.

2.5 Exercise 2.5: A positive real fractional power

What we must show. Prove that $(3 + \sqrt{2})^{2/3}$ is irrational.

Plan. First explain the fractional exponent using the positive real cube root. Cubing the proposed number then produces a simple expression $11 + 6\sqrt{2}$, from which $\sqrt{2}$ can be isolated.

1. Interpret the exponent without a sign ambiguity

Since $\sqrt{2} > 0$, the base

$$u = 3 + \sqrt{2}$$

is positive. Let $t = \sqrt[3]{u} > 0$. Thus $t^3 = u$, and the positive real fractional power in the problem is

$$a = u^{2/3} = (\sqrt[3]{u})^2 = t^2.$$

It follows from integer power rules that

$$a^3 = (t^2)^3 = t^6 = (t^3)^2 = u^2 = (3 + \sqrt{2})^2.$$

This explains precisely why cubing removes the exponent $2/3$.

2. Expand the square completely

Using $(v + w)^2 = v^2 + 2vw + w^2$ gives

$$\begin{aligned} a^3 &= (3 + \sqrt{2})^2 \\ &= 3^2 + 2 \cdot 3 \cdot \sqrt{2} + (\sqrt{2})^2 \\ &= 9 + 6\sqrt{2} + 2 \\ &= 11 + 6\sqrt{2}. \end{aligned}$$

Subtracting 11 and dividing by 6 therefore yields

$$a^3 - 11 = 6\sqrt{2}, \qquad \sqrt{2} = \frac{a^3 - 11}{6}.$$

3. Use rational closure to obtain the contradiction

Suppose $a \in \mathbb{Q}$. Then $a^3 \in \mathbb{Q}$, hence $a^3 - 11 \in \mathbb{Q}$. Because 6 is a nonzero rational integer, division by 6 preserves rationality. The last displayed equation would imply $\sqrt{2} \in \mathbb{Q}$. This contradicts the auxiliary irrationality result in Exercise 2.1.

Conclusion. $(3 + \sqrt{2})^{2/3}$, with the positive real interpretation of the fractional power, is irrational.

2.6 Exercise 2.6: A rational quadratic expression

What we must show. Explain why $b \in \mathbb{Q}$ implies $4 - 7b^2 \in \mathbb{Q}$.

Plan. Start with the defining fraction for a rational number and write $4 - 7b^2$ as one fraction. Check its numerator and denominator separately.

1. Write the rationality hypothesis explicitly

Assume $b \in \mathbb{Q}$. By definition, there are integers m, n such that

$$b = \frac{m}{n}, \qquad n \neq 0.$$

The numerator m may be positive, negative, or zero. No restriction on the sign of b is needed for this argument.

2. Perform the arithmetic over one denominator

Squaring and multiplying gives

$$b^2 = \frac{m^2}{n^2}, \qquad 7b^2 = \frac{7m^2}{n^2}.$$

As $n \neq 0$, its square n^2 is nonzero, so we may also write $4 = 4n^2/n^2$. Subtracting now gives

$$4 - 7b^2 = \frac{4n^2}{n^2} - \frac{7m^2}{n^2} = \frac{4n^2 - 7m^2}{n^2}.$$

The integer operations of squaring, multiplication, and subtraction show that $4n^2 - 7m^2 \in \mathbb{Z}$. The denominator n^2 is a nonzero integer. The displayed fraction is therefore rational by definition.

3. Explain the connection with Example 6

The same reasoning can be summarized using Tool 1: $b \in \mathbb{Q}$ implies $b^2 \in \mathbb{Q}$, then $7b^2 \in \mathbb{Q}$, and then $4 - 7b^2 \in \mathbb{Q}$. This is the rational-closure step used in Example 6. For the particular number considered there, the radical equation also gives

$$7b^2 = 4 - 2\sqrt{3}, \qquad \text{so} \qquad 4 - 7b^2 = 2\sqrt{3}.$$

If that b were rational, the result just proved would make $2\sqrt{3}$ rational. Dividing by the nonzero rational number 2 would make $\sqrt{3}$ rational, contradicting Exercise 2.1. This shows how the closure statement supplies the needed step in an irrationality proof.

Conclusion. For every rational $b = m/n$ with $n \neq 0$, $4 - 7b^2 = (4n^2 - 7m^2)/n^2 \in \mathbb{Q}$.

2.7 Exercise 2.7: Two radicals that simplify to rational numbers

What we must show. Show that

$$\text{(a)} \quad \sqrt{4+2\sqrt{3}}-\sqrt{3}, \qquad \text{(b)} \quad \sqrt{6+4\sqrt{2}}-\sqrt{2}$$

are rational, and find their values. In each expression the final subtracted square root lies outside the first radical.

Plan. Recognize each radicand as the square of a positive sum. Positivity is essential: for a real v, the principal square root satisfies $\sqrt{v^2}=|v|$. To replace that absolute value by v, we check that $v>0$.

1. Part (a): Expand the proposed positive square root

Consider $v=1+\sqrt{3}$. Because $\sqrt{3}>0$, we have $v>0$. Expanding its square gives

$$\begin{aligned}(1+\sqrt{3})^2 &= 1^2+2\cdot 1\cdot\sqrt{3}+(\sqrt{3})^2\\ &= 1+2\sqrt{3}+3\\ &= 4+2\sqrt{3}.\end{aligned}$$

The radicand is positive, and $1+\sqrt{3}$ is a positive number whose square equals that radicand. It is therefore the principal square root:

$$\sqrt{4+2\sqrt{3}}=1+\sqrt{3}.$$

One can also justify the choice of sign from uniqueness. If two nonnegative numbers v, w have the same positive square, then $(v-w)(v+w)=v^2-w^2=0$. Since $v+w>0$, it follows that $v-w=0$, so $v=w$. There is thus only one nonnegative square root here.

Substituting the identity into the expression in part (a) gives

$$\sqrt{4+2\sqrt{3}}-\sqrt{3}=(1+\sqrt{3})-\sqrt{3}=1+(\sqrt{3}-\sqrt{3})=1+0=1.$$

As $1=1/1$, this value is rational.

2. Part (b): Expand the second proposed positive square root

Now consider $w=2+\sqrt{2}$. It is positive because $\sqrt{2}>0$. Its square expands as

$$\begin{aligned}(2+\sqrt{2})^2 &= 2^2+2\cdot 2\cdot\sqrt{2}+(\sqrt{2})^2\\ &= 4+4\sqrt{2}+2\\ &= 6+4\sqrt{2}.\end{aligned}$$

Thus $2+\sqrt{2}$ is the positive square root of $6+4\sqrt{2}$. The sign check permits the identity

$$\sqrt{6+4\sqrt{2}}=2+\sqrt{2}.$$

It follows that

$$\sqrt{6+4\sqrt{2}}-\sqrt{2}=(2+\sqrt{2})-\sqrt{2}=2+(\sqrt{2}-\sqrt{2})=2+0=2.$$

Since $2=2/1$, this value is rational as well.

Conclusion. Part (a) equals 1 and part (b) equals 2; both expressions are rational numbers.

2.8 Exercise 2.8: All rational zeros of an eighth-degree polynomial

What we must show. Find every rational solution of

$$x^8 - 4x^5 + 13x^3 - 7x + 1 = 0.$$

Plan. The polynomial is monic, has integer coefficients, and has constant term 1. The monic rational-zero result restricts every rational root to an integer divisor of 1. We must test both candidates and explain why this exhausts the rational possibilities.

1. Obtain the complete candidate list

Let

$$P(x) = x^8 - 4x^5 + 13x^3 - 7x + 1.$$

The coefficient of x^8 is 1, so P is monic. Its missing powers have coefficient 0, which is also an integer. Its constant coefficient is $1 \neq 0$. Thus Tool 3 applies to every rational zero of P. It says that such a zero must be an integer dividing 1. If an integer m divides 1, then $1 = mk$ for some integer k. Taking absolute values gives $1 = |m|\,|k|$, so both nonnegative integer factors must equal 1. In particular $|m| = 1$, and hence $m = 1$ or $m = -1$. The full rational candidate list is therefore

$$\{1, -1\}.$$

Here is the denominator step directly for this particular polynomial. If a rational root is m/n in lowest terms, with $n > 0$, multiplication of $P(m/n) = 0$ by n^8 gives

$$m^8 - 4m^5n^3 + 13m^3n^5 - 7mn^7 + n^8 = 0.$$

Solving for m^8 gives

$$m^8 = n(4m^5n^2 - 13m^3n^4 + 7mn^6 - n^7),$$

so n divides m^8. Since m, n are relatively prime, repeated coprime cancellation gives $n \mid 1$, hence $n = 1$. After this reduction, $P(m) = 0$ can be rearranged as

$$1 = -m^8 + 4m^5 - 13m^3 + 7m = m(-m^7 + 4m^4 - 13m^2 + 7).$$

This proves $m \mid 1$ and recovers precisely the two candidates above. Also $P(0) = 1$, so zero has not been omitted as a possible root.

2. Evaluate the candidate 1

Every positive power of 1 equals 1. Consequently

$$\begin{aligned} P(1) &= 1^8 - 4 \cdot 1^5 + 13 \cdot 1^3 - 7 \cdot 1 + 1 \\ &= 1 - 4 + 13 - 7 + 1 \\ &= -3 + 13 - 7 + 1 \\ &= 10 - 7 + 1 \\ &= 3 + 1 = 4. \end{aligned}$$

Since $4 \neq 0$, the number 1 is not a solution.

3. Evaluate the candidate -1

The eighth power of -1 is 1, while its fifth and third powers are -1. Substituting with the signs retained gives

$$\begin{aligned} P(-1) &= (-1)^8 - 4(-1)^5 + 13(-1)^3 - 7(-1) + 1 \\ &= 1 - 4(-1) + 13(-1) + 7 + 1 \\ &= 1 + 4 - 13 + 7 + 1 \\ &= 5 - 13 + 7 + 1 \\ &= -8 + 7 + 1 \\ &= -1 + 1 = 0. \end{aligned}$$

Thus -1 is a solution, and it is rational because $-1 = (-1)/1$.

4. Explain why the search is complete

Every rational solution must belong to the candidate set $\{1, -1\}$ from Step 1. Step 2 rejects 1, and Step 3 verifies -1. There are no other rational candidates to examine.

Conclusion. The set of all rational solutions is $\{-1\}$.

3 The set of real numbers

This section concerns the rules of arithmetic, inequalities, and absolute values. We work with the usual operations on real numbers. In particular, *an inverse must belong to the set under discussion*: an integer does not have a multiplicative inverse *in the integers* merely because it has one in the real numbers.

A reference sheet for the rules

Ross's labels will be useful when explaining individual steps. For real numbers x, y, z, the field rules are

$$\begin{aligned} &\text{A1}: \quad x+(y+z)=(x+y)+z, && \text{A2}: \quad x+y=y+x,\\ &\text{A3}: \quad x+0=x, && \text{A4}: \quad x+(-x)=0,\\ &\text{M1}: \quad x(yz)=(xy)z, && \text{M2}: \quad xy=yx,\\ &\text{M3}: \quad x\cdot 1=x, && \text{M4}: \quad xx^{-1}=1 \quad (x\neq 0),\\ &\text{DL}: \quad x(y+z)=xy+xz. \end{aligned}$$

The additive identity 0, the multiplicative identity 1, the additive inverse $-x$, and (when $x \neq 0$) the multiplicative inverse x^{-1} must be elements of the field. A field has more than one element.

The order rules say that any two real numbers can be compared (O1), $x \leq y$ and $y \leq x$ together imply $x = y$ (O2), and $x \leq y \leq z$ implies $x \leq z$ (O3). They also give

$$\text{O4}: \quad x \leq y \implies x+z \leq y+z, \qquad \text{O5}: \quad x \leq y,\ 0 \leq z \implies xz \leq yz.$$

By definition, $x < y$ means that $x \leq y$ and $x \neq y$. Subtraction is addition of the additive inverse: $x - y = x + (-y)$. Division by a nonzero number is multiplication by its multiplicative inverse: $x/y = xy^{-1}$.

We may use the results already established in the text before these exercises, including additive cancellation, $x \cdot 0 = 0$, $(-x)y = -(xy)$, nonnegativity of squares, and the triangle inequality. When an exercise asks for a missing part of a theorem, that missing part is proved below rather than assumed.

3.1 Exercise 3.1: Which rules fail for natural numbers and integers?

What we must show. Determine which of A1–A4, M1–M4, DL, and O1–O5 are valid when the operations and order are the usual ones and all required elements must lie in the specified set.

(a), Step 1: fix the convention for natural numbers.

Ross uses $\mathbb{N} = \{1, 2, 3, \ldots\}$. Thus $1 \in \mathbb{N}$ but $0 \notin \mathbb{N}$. This distinction matters for the identity requirement A3.

(a), Step 2: test the additive identity.

A3 requires an additive identity *in the set*. The usual identity is 0, which is not in $\mathbb{N}$. There is no replacement positive integer e: if $e \in \mathbb{N}$, then $e \geq 1$, so

$$1 + e \geq 1 + 1 = 2 > 1.$$

Consequently $1 + e \neq 1$. Hence no natural number can act as an additive identity, and the internal identity requirement A3 fails.

There is a small interpretive issue here. The equality $n + 0 = n$ is true if we calculate in $\mathbb{Z}$ or $\mathbb{R}$. It is not a false numerical equation. It fails as a *field axiom for* $\mathbb{N}$ because its required constant 0 is outside $\mathbb{N}$. We use this internal interpretation in the answer.

(a), Step 3: test additive inverses.

A4 would require an element of $\mathbb{N}$ that can be added to 1 to give 0. But if $m \in \mathbb{N}$, then $m \geq 1$, and therefore $1 + m \geq 2$. In particular, $1 + m \neq 0$. The additive inverse -1 exists in $\mathbb{Z}$ and $\mathbb{R}$, but not in $\mathbb{N}$. Thus A4 fails.

(a), Step 4: test multiplicative inverses.

M4 would require an element $m \in \mathbb{N}$ with $2m = 1$. Every such m satisfies $m \geq 1$, so $2m \geq 2$. Therefore $2m \neq 1$. Equivalently, the required inverse $1/2$ is not a natural number. Thus M4 fails. It is enough to find the single nonzero element 2 for which the requirement fails; M4 must hold for *every* nonzero element.

(a), Step 5: check the remaining rules.

Addition and multiplication of positive integers again give positive integers. Associativity and commutativity remain true, so A1, A2, M1, and M2 hold. Since $1 \in \mathbb{N}$ and $n \cdot 1 = n$, M3 holds. The distributive identity holds for positive integers, so DL holds. The usual comparisons of natural numbers satisfy O1–O4. Finally, multiplying an inequality between natural numbers by a natural number, which is positive, preserves its direction; this verifies the numerical inequality in O5. The occurrence of 0 in the condition of O5 is understood in the ambient usual order. No additional order inequality becomes false upon restricting its variables to $\mathbb{N}$.

Conclusion. For $\mathbb{N}$ the failed internal field requirements are $\boxed{\text{A3, A4, M4}}$. The remaining listed arithmetic identities and order implications are valid with the usual arithmetic and order.

(b), Step 1: identify what changes for integers.

Now the set is $\mathbb{Z}$, so $0 \in \mathbb{Z}$ and $-n \in \mathbb{Z}$ whenever $n \in \mathbb{Z}$. Thus $n + 0 = n$ takes place inside the set, and $n + (-n) = 0$ also takes place inside the set. A3 and A4 both hold.

(b), Step 2: exhibit the remaining failure.

The element 2 is a nonzero integer. If it had an inverse $m \in \mathbb{Z}$, we would have $2m = 1$. The left side is an even integer, since it is twice an integer; the right side is odd. Hence no such integer m exists. This again says that $1/2 \notin \mathbb{Z}$, so M4 fails.

(b), Step 3: check every other requirement.

The sum and product of integers are integers. The associative, commutative, and distributive identities remain valid. Both identities, 0 and 1, are integers. Additive inverses are integers. The usual order of integers satisfies O1–O4, and multiplication by a nonnegative integer preserves a weak inequality, as required by O5.

Conclusion. For $\mathbb{Z}$ the only failed listed requirement is $\boxed{\text{M4}}$. Some integers, namely 1 and -1, do have integer multiplicative inverses, but that does not meet a requirement imposed on every nonzero integer.

3.2 Exercise 3.2: Locate the hidden uses of additive commutativity

What we must show. Find the uses of A2 in the proofs of Theorem 3.1(ii) and (iii).

Plan. The cancellation rule proved in Theorem 3.1(i) is written with the common term on the *right*:

$$x + c = y + c \quad \Longrightarrow \quad x = y.$$

Before using this particular form, both equalities must have that shape.

(a), Step 1: write what distributivity actually gives.

Set $t = a \cdot 0$. Since $0 + 0 = 0$ by A3, distributivity gives

$$t = a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0 = t + t.$$

To use right cancellation with the common term t, the proof rewrites the leftmost t as $0 + t$.

(a), Step 2: justify that rewriting.

A3 gives $t + 0 = t$, with the zero on the *right*. A2 is needed to move it to the left:

$$0 + t = t + 0 = t.$$

Thus $t = t + t$ becomes $0 + t = t + t$. Right cancellation of t now gives $0 = t = a \cdot 0$.

Conclusion. The hidden use of A2 is the passage from $a \cdot 0$ to $0 + a \cdot 0$: first A3 permits $a \cdot 0 + 0$, then A2 switches the two summands.

(b), Step 1: identify the equality just before cancellation.

The proof obtains

$$ab + (-a)b = ab + \big(-(ab)\big).$$

Here the common summand ab is on the *left*, whereas the previously proved cancellation statement has the common summand on the right.

(b), Step 2: apply A2 to both sums.

The two relevant switches are

$$ab + (-a)b = (-a)b + ab, \qquad ab + \big(-(ab)\big) = \big(-(ab)\big) + ab.$$

The equality therefore becomes

$$(-a)b + ab = \big(-(ab)\big) + ab.$$

Now right cancellation with common term ab proves $(-a)b = -(ab)$.

(b), Step 3: distinguish addition from multiplication.

There are also commutative switches in the preceding factorization, but those use M2, not A2. For example,

$$\begin{aligned} ab + (-a)b &= ba + b(-a) && \text{by M2 in the two products,} \\ &= b(a + (-a)) && \text{by DL, read in reverse,} \\ &= b \cdot 0 && \text{by A4,} \end{aligned}$$

$$= 0 \qquad \text{by Theorem 3.1(ii).}$$

No exchange of *summands* occurs in this displayed chain.

Conclusion. The A2 needed for the final use of Theorem 3.1(i) moves the common additive term ab from the left to the right on both sides. Product switches elsewhere in the argument use M2.

3.3 Exercise 3.3: A product of two negatives and multiplicative cancellation

What we must show. Prove Theorem 3.1(iv), $(-a)(-b) = ab$, and Theorem 3.1(v), $ac = bc$ with $c \neq 0$ implies $a = b$.

Part (iv), Step 1: prove that taking the negative twice returns the number.

For any real u, A4 gives

$$u + (-u) = 0.$$

Apply A4 again, now to the number $-u$:

$$(-u) + \big(-(-u)\big) = 0.$$

By A2 the second left side is $-(-u) + (-u)$. Hence

$$u + (-u) = -(-u) + (-u).$$

The common right summand is $-u$. Additive cancellation gives

$$u = -(-u).$$

This proves the double-negative rule without using a product of two negative numbers.

Part (iv), Step 2: reduce the product to the rule already proved.

Theorem 3.1(iii) states $(-x)y = -(xy)$ for all real x, y. First apply it with $x = a$ and $y = -b$:

$$(-a)(-b) = -\big(a(-b)\big).$$

Next rewrite the product inside the parentheses:

$$\begin{aligned} a(-b) &= (-b)a && \text{by M2,} \\ &= -(ba) && \text{by Theorem 3.1(iii) with } x = b,\ y = a, \\ &= -(ab) && \text{by M2 inside the negative.} \end{aligned}$$

Substitute this expression into the previous equality:

$$(-a)(-b) = -\big(-(ab)\big) = ab.$$

The last equality is Step 1 with $u = ab$.

Conclusion. For all real a, b, $\boxed{(-a)(-b) = ab}$.

Part (v), Step 1: use the nonzero assumption.

Suppose $ac = bc$ and $c \neq 0$. By M4 the number c^{-1} exists and satisfies $cc^{-1} = 1$. Multiply both sides of the given equality on the right by this *same* real number:

$$(ac)c^{-1} = (bc)c^{-1}.$$

Part (v), Step 2: simplify each side, naming each rule.

For the left side,

$$(ac)c^{-1} = a(cc^{-1}) = a \cdot 1 = a.$$

These equalities use M1, M4, and M3, respectively. The same rules give

$$(bc)c^{-1} = b(cc^{-1}) = b \cdot 1 = b.$$

Replacing the two sides of $(ac)c^{-1} = (bc)c^{-1}$ by these simplified forms gives $a = b$.

Conclusion. If $ac = bc$ and $c \neq 0$, then $\boxed{a = b}$. The condition $c \neq 0$ is necessary: $1 \cdot 0 = 2 \cdot 0$, but $1 \neq 2$.

3.4 Exercise 3.4: Positivity of one and reversal of reciprocals

What we must show. Prove Theorem 3.2(v), $0 < 1$, and Theorem 3.2(vii):

$$0 < a < b \quad \Longrightarrow \quad 0 < b^{-1} < a^{-1}.$$

Part (v), Step 1: show that 1 is nonnegative.

Theorem 3.2(iv), already proved in the text, says $0 \leq x^2$ for every real x. Take $x = 1$. By M3, $1^2 = 1 \cdot 1 = 1$. Therefore

$$0 \leq 1.$$

Part (v), Step 2: rule out equality with zero.

Suppose, for a contradiction, that $1 = 0$. Let x be any element of the field. Then

$$x = x \cdot 1 = x \cdot 0 = 0.$$

The first equality uses M3; the second substitutes $1 = 0$; the third uses Theorem 3.1(ii). Thus *every* field element would equal 0, leaving only one element. This contradicts the requirement that a field have more than one element. Consequently $1 \neq 0$.

Conclusion. Since $0 \leq 1$ and $0 \neq 1$, the definition of strict inequality gives $\boxed{0 < 1}$.

Part (vii), Step 1: record why the inverses are positive.

If $t > 0$, then $t \neq 0$, so M4 supplies t^{-1} with $tt^{-1} = 1$. Here is the proof of its positivity, as in Theorem 3.2(vi). First t^{-1} cannot be zero, because then $1 = tt^{-1} = t \cdot 0 = 0$, contradicting Part (v). If $t^{-1} < 0$, order reversal under negation gives $0 < -t^{-1}$. The product of nonnegative numbers is nonnegative (Theorem 3.2(iii)), so

$$0 \leq t(-t^{-1}) = -(tt^{-1}) = -1.$$

The product identity in the middle follows from Theorem 3.1(iii) and M2. Negating $0 \leq -1$ reverses its direction and gives $1 \leq 0$. Together with $0 \leq 1$, O2 would give $1 = 0$, again contradicting Part (v). Thus t^{-1} is neither zero nor negative, so $t^{-1} > 0$.

In the hypothesis $0 < a < b$, we have $a > 0$ and $b > 0$. Applying the preceding argument to $t = a$ and $t = b$ gives

$$a^{-1} > 0, \qquad b^{-1} > 0.$$

Part (vii), Step 2: justify multiplication of strict inequalities.

If $x < y$ and $c > 0$, then O5 first gives the weak inequality $xc \leq yc$. Equality $xc = yc$ would imply $x = y$ by Exercise 3.3(v), since $c \neq 0$. But $x < y$ includes $x \neq y$. Therefore equality is impossible, and

$$x < y, \quad c > 0 \quad \Longrightarrow \quad xc < yc.$$

This explains why strictness is retained in the following two steps.

Part (vii), Step 3: multiply first by a^{-1}.

Start from $a < b$ and use the positive multiplier a^{-1}:

$$aa^{-1} < ba^{-1}.$$

M4 turns the left side into 1, so

$$1 < ba^{-1}.$$

Part (vii), Step 4: multiply next by b^{-1}.

The multiplier b^{-1} is also positive. Thus

$$1 \cdot b^{-1} < (ba^{-1})b^{-1}.$$

The left side equals b^{-1} by M2 and M3. On the right,

$$(ba^{-1})b^{-1} = (a^{-1}b)b^{-1} = a^{-1}(bb^{-1}) = a^{-1} \cdot 1 = a^{-1}.$$

These steps use M2, M1, M4, and M3, in that order. Hence $b^{-1} < a^{-1}$.

Conclusion. Combining this strict inequality with $b^{-1} > 0$ gives $\boxed{0 < b^{-1} < a^{-1}}$. Positivity of a and b is essential to the choice of positive multipliers.

3.5 Exercise 3.5: Absolute-value bounds and the reverse triangle inequality

What we must show. Prove (a) $|b| \leq a$ if and only if $-a \leq b \leq a$, and (b) $||a| - |b|| \leq |a - b|$.

Part (a), Step 1: unpack the definition of absolute value.

For a real number x,

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

In both cases $|x| \geq 0$. We also have

$$-|x| \leq x \leq |x|.$$

To verify the second statement rather than assume it, consider the two cases. If $x \geq 0$, then $|x| = x$, the upper inequality is equality, and the lower inequality is $-x \leq x$, since $-x \leq 0 \leq x$. If $x < 0$, then $|x| = -x$, the lower inequality is equality $-|x| = x$, and the upper inequality follows from $x < 0 < -x$.

Part (a), Step 2: prove the forward implication.

Assume $|b| \leq a$. The bounds from Step 1 give $b \leq |b|$. Transitivity then gives

$$b \leq |b| \leq a,$$

so $b \leq a$. Negating the assumed inequality reverses its direction:

$$-a \leq -|b|.$$

Since $-|b| \leq b$ by Step 1, transitivity gives $-a \leq b$. The two conclusions together are $-a \leq b \leq a$.

Part (a), Step 3: prove the reverse implication.

Assume both $-a \leq b$ and $b \leq a$. If $b \geq 0$, then $|b| = b$, so the second assumption gives $|b| \leq a$. If $b < 0$, negate the first assumption $-a \leq b$ to obtain $-b \leq a$. In this case $|b| = -b$, so again $|b| \leq a$. These two cases exhaust all real b by O1.

Part (a), Step 4: check zero and negative bounds.

If $a = 0$, the left condition says $|b| \leq 0$, which forces $|b| = 0$ and hence $b = 0$ by the definition. The right condition is $0 \leq b \leq 0$, which forces $b = 0$ by O2. If $a < 0$, the left condition is impossible because $0 \leq |b|$, whereas the right condition is impossible because $-a > 0 > a$: no real b can simultaneously be at least $-a$ and at most a. Thus the equivalence includes these bounds as well.

Conclusion. For all real a, b, including $a \leq 0$, $\boxed{|b| \leq a \iff -a \leq b \leq a}$.

Part (b), Step 1: obtain a one-sided bound from the triangle inequality.

The equality $a = (a - b) + b$ follows by associating $a + ((-b) + b)$ and using $(-b) + b = 0$. Apply the triangle inequality to the sum $(a - b) + b$:

$$|a| = |(a - b) + b| \leq |a - b| + |b|.$$

Add $-|b|$ to both sides using O4. After associating and cancelling $|b| + (-|b|)$, the result is

$$|a| - |b| \leq |a - b|. \tag{1}$$

Part (b), Step 2: bound the negative of the same expression.

Interchange the roles of a and b in the preceding argument:

$$|b| - |a| \leq |b - a|. \tag{2}$$

We need to identify the right side. First, $b - a = -(a - b)$, as follows by distributing the negative over $a + (-b)$ and using the double-negative rule. Also $|-x| = |x|$ for any real x: when $x > 0$, the two absolute values are $-(-x) = x$ and x; when $x < 0$, both are $-x$; when $x = 0$, both are 0. Therefore

$$|b - a| = |-(a - b)| = |a - b|.$$

Substitute this into (2):

$$|b| - |a| \leq |a - b| .$$

Negating both sides reverses the inequality, so

$$-|a - b| \leq |a| - |b| . \tag{3}$$

Part (b), Step 3: combine and apply part (a).

Together (1) and (3) give the two-sided bound

$$-|a - b| \leq |a| - |b| \leq |a - b| .$$

In part (a), take its bounded number to be $|a| - |b|$ and its bound to be $|a - b|$. That equivalence turns this two-sided bound into

$$||a| - |b|| \leq |a - b| .$$

Conclusion. The reverse triangle inequality is $\boxed{||a| - |b|| \leq |a - b|}$ for every real a, b.

3.6 Exercise 3.6: The triangle inequality for three and for finitely many terms

What we must show. Prove (a) $|a + b + c| \leq |a| + |b| + |c|$ and (b) $|a_1 + \cdots + a_n| \leq |a_1| + \cdots + |a_n|$ for every positive integer n and every choice of real summands.

Part (a), Step 1: group the sum into two terms.

By A1, $a + b + c = (a + b) + c$. The two-term triangle inequality, applied to the real numbers $a + b$ and c, gives

$$|a + b + c| \leq |a + b| + |c| . \tag{4}$$

Part (a), Step 2: apply the two-term inequality again.

Apply it to a and b:

$$|a + b| \leq |a| + |b| .$$

Add $|c|$ to both sides by O4:

$$|a + b| + |c| \leq |a| + |b| + |c| . \tag{5}$$

By transitivity, combining (4) with (5) gives

$$|a + b + c| \leq |a| + |b| + |c| .$$

Conclusion. The three-term inequality follows from two applications of the two-term inequality; no enumeration of the signs of a, b, c is needed.

Part (b), Step 1: state exactly what is being proved by induction.

Let P_n be the assertion that

$$|a_1 + \cdots + a_n| \leq |a_1| + \cdots + |a_n|$$

for *every* list of n real numbers $a_1, \ldots, a_n$. This quantifier is important: in the induction step we may choose any list of $n + 1$ numbers and apply P_n to its first n entries.

Part (b), Step 2: verify the base case.

For $n = 1$, the claimed inequality is simply $|a_1| \le |a_1|$. The two sides are equal, so P_1 holds for every real a_1.

Part (b), Step 3: state the induction hypothesis and the next target.

Fix $n \ge 1$ and assume P_n. Let $a_1, \dots, a_n, a_{n+1}$ be arbitrary real numbers. The target is

$$|a_1 + \cdots + a_n + a_{n+1}| \le |a_1| + \cdots + |a_n| + |a_{n+1}| \,.$$

Set $s = a_1 + \cdots + a_n$. The induction hypothesis gives $|s| \le \sum_{j=1}^{n} |a_j|$.

Part (b), Step 4: prove the induction step.

The two-term triangle inequality and then the induction hypothesis give

$$\begin{aligned} |a_1 + \cdots + a_n + a_{n+1}| &= |s + a_{n+1}| && \text{by the definition of } s \text{ and A1,} \\ &\le |s| + |a_{n+1}| && \text{by the triangle inequality,} \\ &\le \sum_{j=1}^{n} |a_j| + |a_{n+1}| && \text{by the induction hypothesis and O4,} \\ &= \sum_{j=1}^{n+1} |a_j| && \text{by the definition of the sum.} \end{aligned}$$

The list of $n + 1$ real numbers was arbitrary, so P_{n+1} follows.

Conclusion. Since P_1 holds and P_n implies P_{n+1} for every $n \ge 1$, mathematical induction proves the finite-sum triangle inequality for every $n \in \mathbb{N}$.

3.7 Exercise 3.7: Strict absolute-value bounds and intervals centered at a point

What we must show. Establish all three equivalences:

$$\begin{array}{lll} \text{(a)} & |b| < a & \iff \quad -a < b < a, \\ \text{(b)} & |a - b| < c & \iff \quad b - c < a < b + c, \\ \text{(c)} & |a - b| \le c & \iff \quad b - c \le a \le b + c. \end{array}$$

Part (a), Step 1: prove the forward implication.

Suppose $|b| < a$. From Exercise 3.5(a), Step 1, we know $b \le |b|$ and $-|b| \le b$. Thus

$$b \le |b| < a \quad \Longrightarrow \quad b < a.$$

Negating $|b| < a$ gives $-a < -|b|$. This strict reversal is justified by Theorem 3.2(i) for the weak inequality; equality of the negatives would imply equality of the originals by taking their negatives again, so equality is excluded. Combining it with $-|b| \le b$ gives $-a < b$. Together these conclusions are $-a < b < a$.

Part (a), Step 2: prove the reverse implication.

Assume $-a < b < a$. If $b \geq 0$, then $|b| = b < a$ by the right-hand assumed inequality. If $b < 0$, negate the left-hand assumed inequality $-a < b$ to get $-b < a$; since $|b| = -b$ in this case, we again obtain $|b| < a$. The case $b = 0$ belongs to the first case.

Part (a), Step 3: check nonpositive values of the bound.

If $a \leq 0$, the left condition cannot hold: it would give $0 \leq |b| < a \leq 0$. The right condition cannot hold either. Indeed $a \leq 0$ implies $0 \leq -a$, hence $a \leq -a$, while $-a < b < a$ would imply $-a < a$. Those inequalities contradict each other. Thus the equivalence remains true when both conditions are false.

Conclusion. For all real a, b, $\boxed{|b| < a \iff -a < b < a}$. Either condition, when true, forces $a > 0$.

Part (b), Step 1: prove the forward implication.

Assume $|a - b| < c$. Apply part (a) to the real number $a - b$ with bound c. It gives

$$-c < a - b < c.$$

Add b to the three expressions. Addition preserves strict inequality: O4 preserves the weak inequality, and equality after addition would contradict additive cancellation. Thus

$$-c + b < (a - b) + b < c + b.$$

A1 and A4 give $(a - b) + b = a$; A2 gives $-c + b = b - c$ and $c + b = b + c$. Therefore $b - c < a < b + c$.

Part (b), Step 2: prove the reverse implication.

Assume $b - c < a < b + c$. Add $-b$ to every expression:

$$(b - c) - b < a - b < (b + c) - b.$$

Associativity and commutativity let b and $-b$ become adjacent; A4 and A3 then give $(b - c) - b = -c$ and $(b + c) - b = c$. We obtain $-c < a - b < c$, and part (a) gives $|a - b| < c$.

Conclusion. For all real a, b, c, $\boxed{|a - b| < c \iff b - c < a < b + c}$. If $c \leq 0$, both conditions are false, as follows from part (a); thus no unstated assumption $c > 0$ is needed for the equivalence.

Part (c), Step 1: prove the forward implication.

Assume $|a - b| \leq c$. Exercise 3.5(a), with bounded number $a - b$ and bound c, gives

$$-c \leq a - b \leq c.$$

Use O4 to add b throughout:

$$-c + b \leq (a - b) + b \leq c + b.$$

The same algebra as in part (b) simplifies the three expressions, giving $b - c \leq a \leq b + c$.

Part (c), Step 2: prove the reverse implication.

Assume $b - c \leq a \leq b + c$. Add $-b$ throughout by O4 and simplify:

$$(b - c) - b \leq a - b \leq (b + c) - b \quad \Longrightarrow \quad -c \leq a - b \leq c.$$

Exercise 3.5(a) now gives $|a - b| \leq c$.

Part (c), Step 3: check the boundary cases.

If $c < 0$, an absolute value cannot be at most c, so the left condition is false. Also $b - c > b + c$, since subtracting gives $(b - c) - (b + c) = -2c > 0$; hence no a can lie between those endpoints in the specified order. The right condition is false as well. If $c = 0$, the left condition forces $a - b = 0$, or $a = b$; the right condition becomes $b \le a \le b$, which also forces $a = b$ by O2.

Conclusion. For all real a, b, c, $\boxed{|a - b| \le c \iff b - c \le a \le b + c}$.

3.8 Exercise 3.8: Comparison with every number above a fixed bound

What we must show. Suppose that $a, b \in \mathbb{R}$ and

$$a \le b_1 \qquad \text{for every real number } b_1 \text{ satisfying } b_1 > b. \tag{6}$$

Prove $a \le b$.

Plan. If a were above b, the arithmetic midpoint between a and b would be above b but still below a. That specific choice would contradict (6). We will calculate both comparisons explicitly.

Step 1: negate the desired conclusion.

Suppose, for a contradiction, that $a \le b$ is false. By O1, either $a \le b$ or $b \le a$. Since the first alternative is excluded, $b \le a$. Equality $a = b$ would imply $a \le b$, so equality is also excluded. Consequently $b < a$, or equivalently $a > b$. Adding $-b$ to this strict inequality gives

$$d := a - b > 0.$$

Step 2: justify division by two and its sign.

Exercise 3.4 gives $1 > 0$. Adding 1 to $0 < 1$ gives $1 < 2$, and transitivity with $0 < 1$ gives $2 > 0$. Thus $2 \ne 0$, so its inverse $1/2$ exists and is positive by Theorem 3.2(vi). Multiplying $0 < d$ by the positive number $1/2$ preserves strict inequality, by the argument in Exercise 3.4(vii), Step 2. Hence

$$\frac{d}{2} > 0.$$

Step 3: construct a number to which the hypothesis applies.

Define

$$m = \frac{a + b}{2}.$$

We compute its difference from b using a common denominator:

$$\begin{aligned}
m - b &= \frac{a + b}{2} - \frac{2b}{2} && \text{because } 2 \ne 0 \text{ and } (2b)/2 = b, \\
&= \frac{a + b - 2b}{2} && \text{by DL and the definition of division,} \\
&= \frac{a - b}{2} && \text{since } b - 2b = -b, \\
&= \frac{d}{2} > 0 && \text{by the definition of } d \text{ and Step 2.}
\end{aligned}$$

Adding b to $m - b > 0$ gives $m > b$. Thus $b_1 = m$ satisfies the restriction in (6).

Step 4: show that this same number is below a.

Again use a common denominator:

$$\begin{aligned} a - m &= \frac{2a}{2} - \frac{a+b}{2}, \\ &= \frac{2a - a - b}{2}, \\ &= \frac{a-b}{2}, \\ &= \frac{d}{2} > 0. \end{aligned}$$

Adding m throughout $a - m > 0$ gives $a > m$, or $m < a$.

Step 5: apply the universal hypothesis and exhibit the contradiction.

Since $m > b$, (6) applies with $b_1 = m$ and gives $a \leq m$. Step 4 gave $m < a$, which includes $m \leq a$ and $m \neq a$. From $a \leq m$ and $m \leq a$, O2 gives $a = m$, contrary to $m \neq a$. The assumption $a > b$ leads to a contradiction. Therefore $a \leq b$.

Conclusion. The hypothesis forces $\boxed{a \leq b}$. Only ordered-field arithmetic was used: the point m was constructed directly, without a completeness axiom or an appeal to a limit.